

Inverses

Recall: $f: A \rightarrow B$ and $g: B \rightarrow A$ are

inverses if

$$\begin{array}{l} g \circ f = \text{id}_A \\ f \circ g = \text{id}_B \end{array}$$

(*)

Idea: f and g 'undo' each other.

• start w/ f , solve (*) for g

Exs. $\ln, e^x$
 $x^3, x^{1/3}$

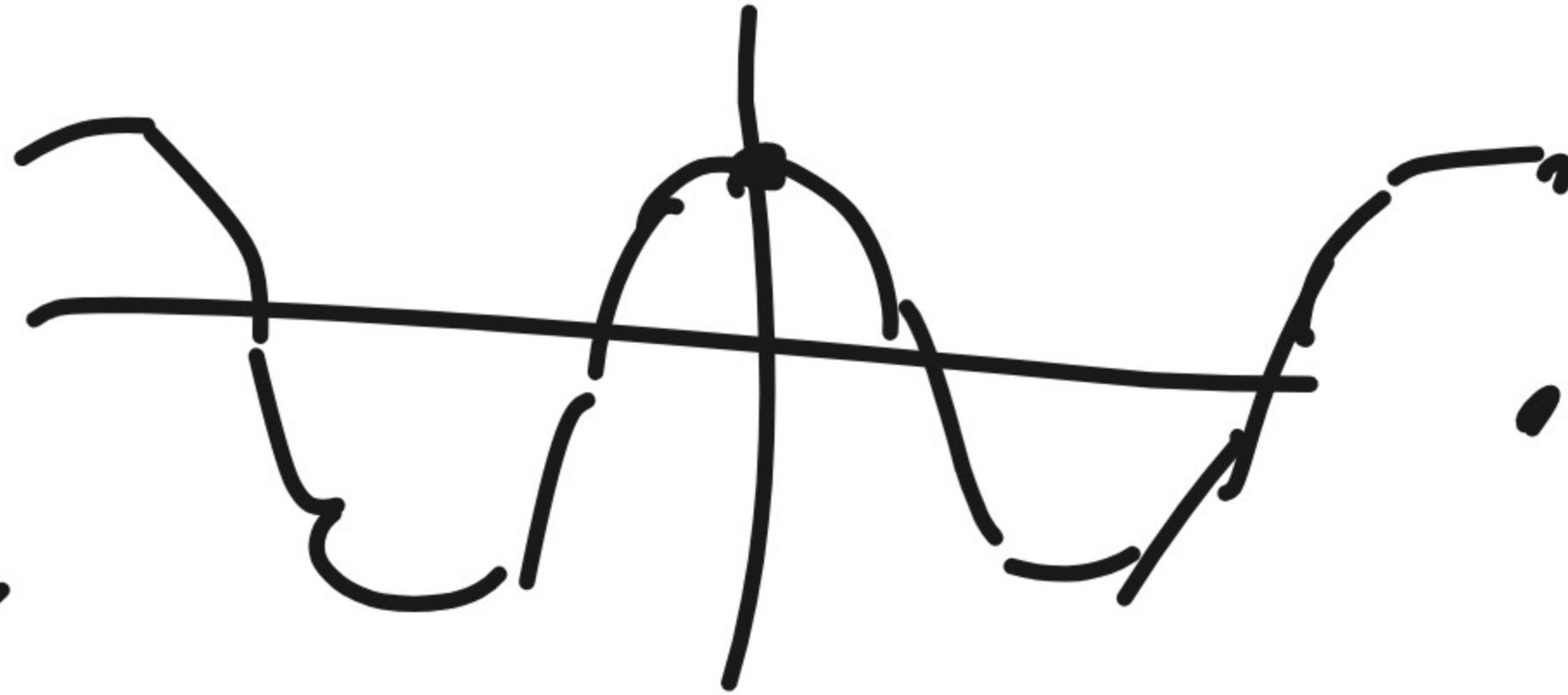
THM: f is invertible $\iff$ f is not inv $\implies$
 f is bijective, f is not bi

N.B. $\Rightarrow$ typical exam problem.

THM in practice

• $f: \mathbb{R} \rightarrow \mathbb{R}$

$x \mapsto \cos x$



not inv. b/c not bi.

Problem: $\cos^{-1}(1) = 0? \quad 2\pi?$

$\cos^{-1}(2) = ?$ No sol to $\cos x = 2$

Ex. $\mathbb{R} \xrightarrow{f} \mathbb{R}$ is bij

$$x \mapsto x^5 + x^3 + x$$

$$f' = 5x^4 + 3x^2 + 1 \geq 1 \quad \forall x \Rightarrow \text{inc} \Rightarrow \text{inj}$$

$$\lim_{x \rightarrow \pm\infty} f = \pm\infty \quad \& \quad f \text{ is cts} \Rightarrow \text{surj (by IVT)}$$

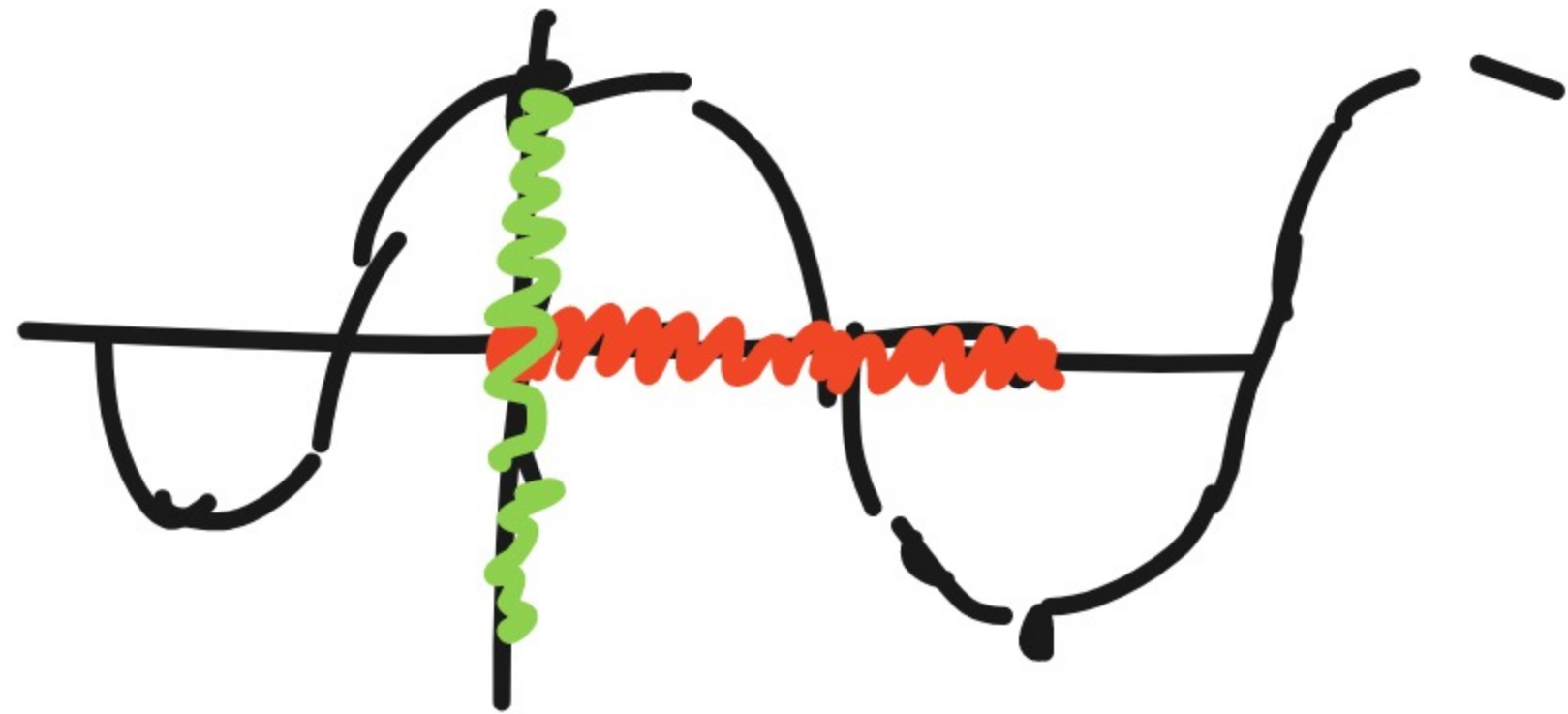
Thus, by thm f is bij.

@TOPH, no formula for f^{-1}

"Fix" cos :

~~$\mathbb{R} \rightarrow \mathbb{R}$~~

No



$$[0, \pi] \rightarrow [-1, 1]$$

Yes

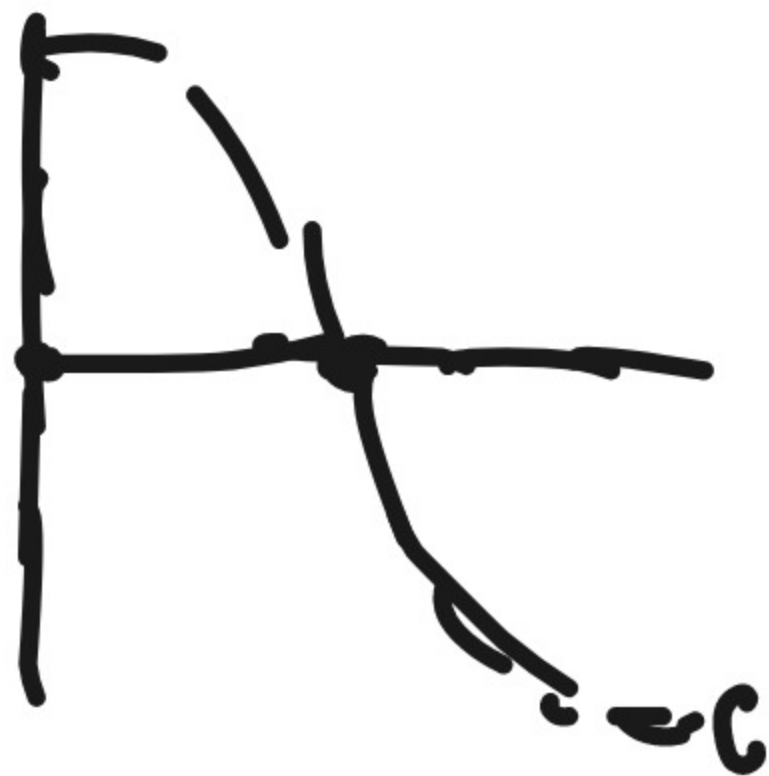
$$x \mapsto \cos x$$

$$[-1, 1] \rightarrow [0, \pi)$$

$$x \mapsto \cos^{-1} x \text{ or } \arccos$$

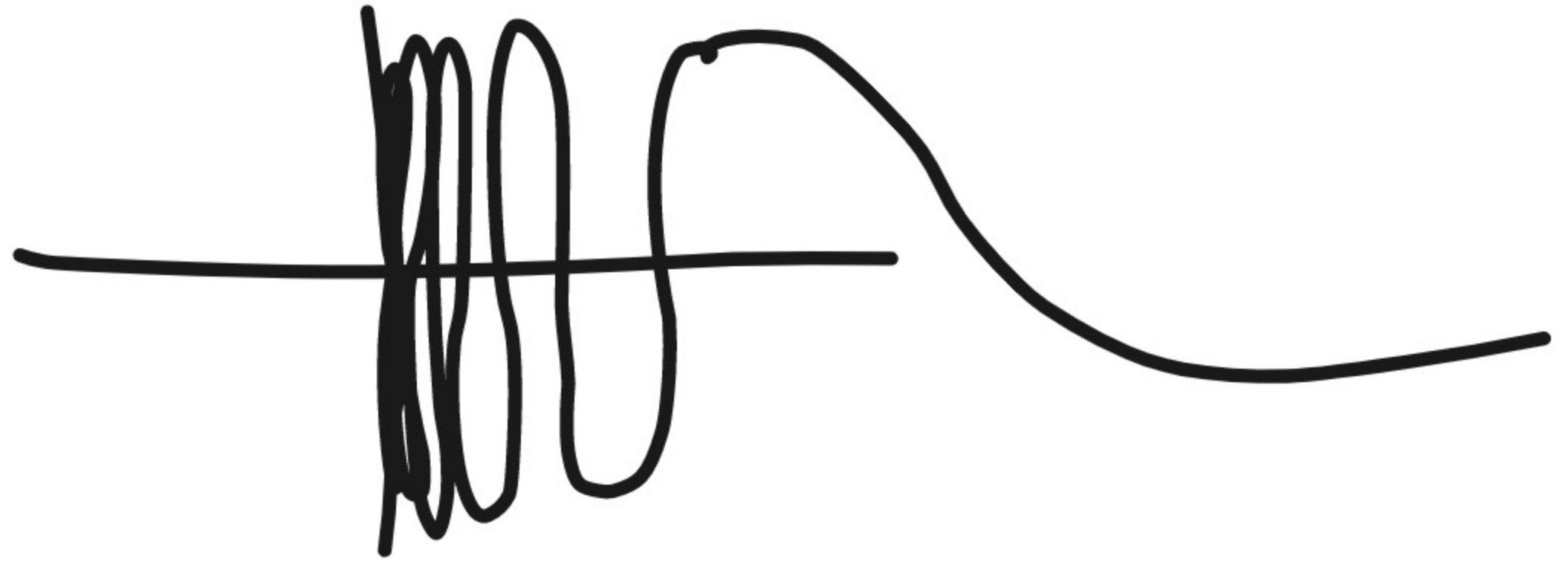
$$\arccos(1) = 0$$

$$\arccos(0) = \pi/2$$



Ex: $\mathbb{R} \rightarrow [-1, 1]$

$$x \mapsto \sin \frac{1}{x}$$



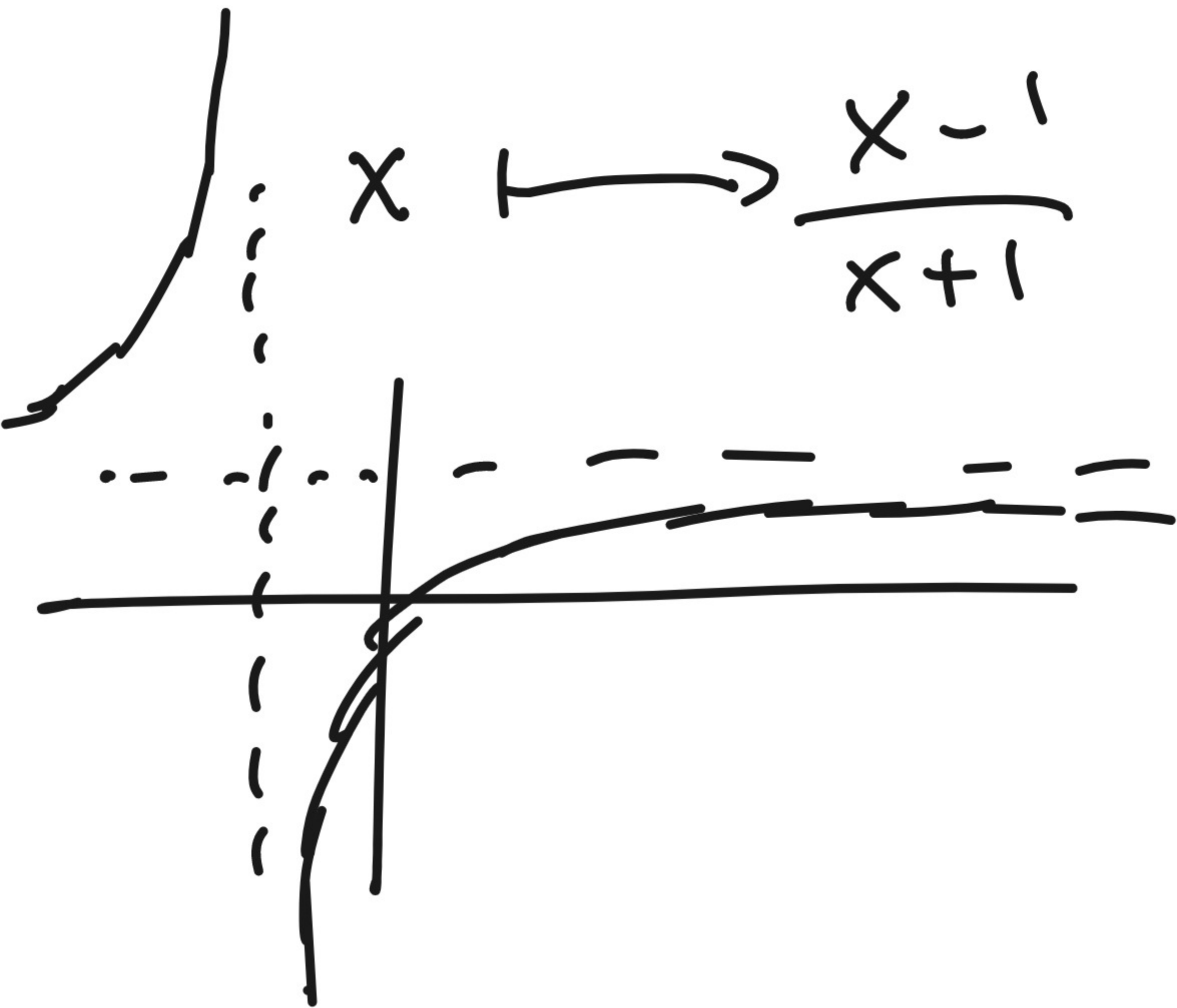
As $x \rightarrow 0^+$, $\sin \frac{1}{x}$ wildly oscillates

but $1 \neq -1$

$\nexists$ Near $x=0$, ^{reasonable} way to make

$\sin \frac{1}{x}$ injective

Ex: $\mathbb{R} - \{-1\} \xrightarrow{f} \mathbb{R} - \{1\}$



Claim: f is bij
(thus invertible).

Warning: f is not cts
@ $x = -1$. (undefined).

OK: Use IVT twice.

Pf: "in". let $a, b \in \mathbb{R} - \{-1\}$. Suppose $f(a) = f(b)$

Then $\frac{a-1}{a+1} = \frac{b-1}{b+1}$. Then $(a-1)(b+1) = (a+1)(b-1)$,

i.e., $\underline{ab - b + a - 1} = \underline{ab + b - a - 1}$.

Thus $-b + a = b - a$, so $2b = 2a$, Thus $a = b$,

and f is $\hat{1}$.

"Surj": $\lim_{x \rightarrow \infty} f = 1 \quad + \quad \lim_{x \rightarrow -1^+} f = -\infty.$

Since f is cts on $(-1, \infty)$, by IVT, $f((-1, \infty)) \supseteq (-\infty, 1).$

Similarly, since $\lim_{x \rightarrow -\infty} f = 1$ and $\lim_{x \rightarrow -1^-} f = \infty$, and

Since f is cts on $(-\infty, -1)$, By IVT, $f((-\infty, -1)) \supseteq (1, \infty).$

But, $f((-\infty, -1) \cup (-1, \infty)) = f((-\infty, -1)) \cup f((-1, \infty)) \supseteq (-\infty, 1) \cup (1, \infty) \stackrel{\mathbb{R}-\{1\}}{=} \mathbb{R}.$

$$\text{domain} = \mathbb{R} - \{1\} = (-\infty, -1) \cup (-1, \infty)$$

$$\text{IVT} : f(-\infty, -1) = (1, \infty)$$

$$f(-1, \infty) = (-\infty, 1)$$

$$\text{Identity: } f(\mathbb{R} - \{1\}) = f(-\infty, -1) \cup (-1, \infty)$$

$$= f(-\infty, -1) \cup f(-1, \infty) \supseteq (1, \infty) \cup (-\infty, 1)$$

$$= \mathbb{R} - \{1\}$$

Sometimes, it is easier to find an inverse
than to prove bijectiv.

$$f(x) = y \quad \Leftrightarrow \quad f^{-1}(y) = x$$

$$f(x) = \frac{x-1}{x+1} = y \quad \Leftrightarrow f^{-1}(y) = x$$

$$\begin{array}{l|l} \begin{array}{l} \text{"} \\ x+1-2 \\ \hline x+1 \\ \text{"} \\ 1 - \frac{2}{x+1} \\ \text{"} \\ y \end{array} & \begin{array}{l} y = 1 - \frac{2}{x+1} \\ y-1 = \frac{-2}{x+1} \\ x+1 = \frac{-2}{y-1} \end{array} \end{array} \quad \begin{array}{l} x = -1 - \frac{2}{y-1} \\ = \frac{-y+1-2}{y-1} = \frac{-y-1}{y-1} \\ = -\frac{(y+1)}{y-1} = \frac{y+1}{1-y} \end{array}$$

$$\mathbb{R} - \{ -1 \} \xrightarrow{f} \mathbb{R} - \{ -1 \}$$

$$x \mapsto \frac{x-1}{x+1}$$

$$\mathbb{R} - \{ -1 \} \xrightarrow{g} \mathbb{R} - \{ -1 \}$$

$$y \mapsto \frac{1+y}{1-y}$$

$$\left\{ \begin{array}{l} (g \circ f)(x) = x \\ (f \circ g)(x) = x \end{array} \right.$$

f has an inverse
 $\Rightarrow f$ is invertible.

How to verify

$$(g \circ f)(x) = x ?$$

$$g(f(x)) = g\left(\frac{x-1}{x+1}\right) = \frac{1 + \left(\frac{x-1}{x+1}\right)}{1 - \left(\frac{x-1}{x+1}\right)} =$$

$$\frac{(x+1) + (x-1)}{\cancel{x+1}}$$

$$\cancel{x+1}$$

$$\frac{\cancel{x+1} - (x-1)}{\cancel{x+1}}$$

$$\cancel{x+1}$$

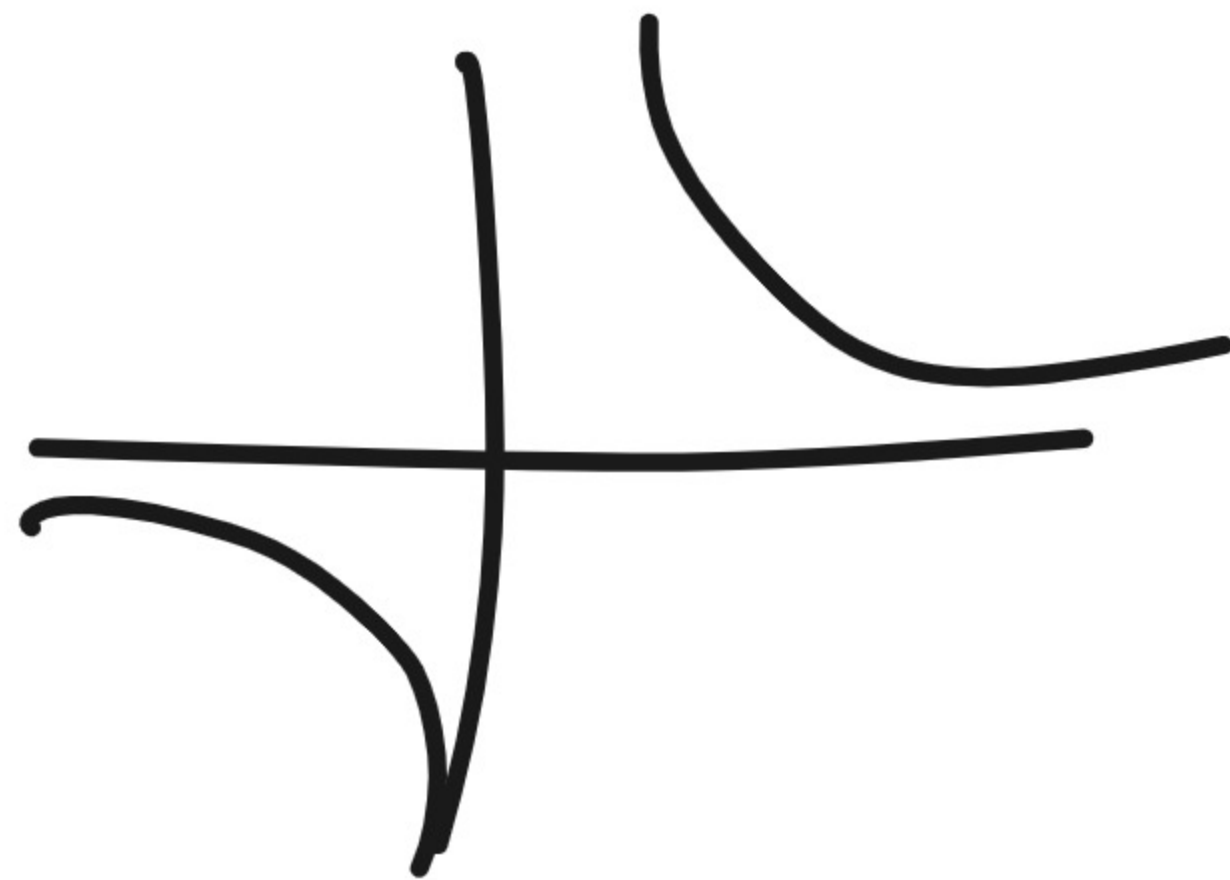
$$=$$

$$\frac{2x}{2} = x$$

log ...

Ex: $\mathbb{R} - \{0\} \xrightarrow{f} \mathbb{R} - \{0\}$

$$x \mapsto \frac{1}{x}$$



f is inverse ... to itself.

$f \circ g = \text{id}$		$\exists E \quad f = g$		$\exists a \text{ sol}$
$g \circ f = \text{id}$		to		



$$(f \circ f)(x) = f(f(x)) = f\left(\frac{1}{x}\right) = \frac{1}{\left(\frac{1}{x}\right)} = x$$

Conclusion:

- $f = f^{-1}$

- f is bijective.

(by THM),

$$= id(x)$$

Ex: $A \xrightarrow{\text{id}} A$

$$\text{id}(x) = x \iff \text{id}^{-1}(x) = x.$$

What is id^{-1} ?

$$\text{id}^{-1} = \text{id}$$

$$\text{id}(x) = x$$

$$\text{id} \circ g = \text{id}_A$$

$$g \circ \text{id} = \text{id}_A$$

$$\text{id} \circ \text{id} = \text{id}$$

$$g = \text{id} \text{ solves}$$

$$\begin{array}{l} \forall f, \\ \text{id} \circ f = f \end{array}$$

Lemma: " $\underbrace{f: A \rightarrow B \quad g, h: B \rightarrow A}_{\text{Inverses are unique}}$, i.e., if g and h are both inverse to f , then $g = h$.

Pf: Since g and h are inverse to f ,

$$g \circ f = \text{id}_A$$

$$f \circ g = \text{id}_B$$

$$\text{and } h \circ f = \text{id}_A$$

$$f \circ h = \text{id}_B.$$

Since $g \circ f = \text{id}_A$, $(g \circ f) \circ h = \text{id}_A \circ h$.

Then $(g \circ f) \circ h = \text{id}_A \circ h = h$.

"
 $g \circ (f \circ h)$
"
 $g \circ \text{id}_B$
"
 g

Thus $g = h$. $\square$

Ex: $\mathbb{R}^2 \xrightarrow{f} \mathbb{R}^2$

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x-1 \\ x+1 \end{pmatrix}$$

$$\mathbb{R}^2 \xrightarrow{g} \mathbb{R}^2$$

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} \frac{x+1}{2} \\ \frac{-x+1}{2} \end{pmatrix}$$

$$g = f^{-1}$$

Pf: $(g \circ f)\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = g\left(f\left(\begin{pmatrix} x \\ y \end{pmatrix}\right)\right) =$

$$g\left(\begin{pmatrix} x-1 \\ x+1 \end{pmatrix}\right) =$$

$$\begin{pmatrix} \frac{(\cancel{x-1}) + (\cancel{x+1})}{2} \\ \frac{-(\cancel{x-1}) + (\cancel{x+1})}{2} \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

($f \circ g$ is similar.)

$$= \text{id}_{\mathbb{R}^2}\left(\begin{pmatrix} x \\ y \end{pmatrix}\right).$$

$\Leftarrow$: if $g = f^{-1}$, then $f = g^{-1}$. $(\exists (f^{-1})^{-1} = f)$

In words: if g "undoes f ", then f undoes g .

P: s.t. $g = f^{-1}$. Then

$$\begin{aligned} g \circ f &= \text{id}_A \\ f \circ g &= \text{id}_B \end{aligned}$$

$$\begin{aligned} f \circ g &= \text{id} \\ g \circ f &= \text{id} \end{aligned}$$

Thus $f = g^{-1}$. $\square$

Started f and solve for g .
if instead we started w g , f solves.

$$\boxed{f(x) = y \stackrel{(*)}{\Leftrightarrow} f^{-1}(y) = x} \quad \left(\begin{array}{l} \text{the input to } f \text{ whose} \\ \text{output is } y \end{array} \right)$$

Sup $f \circ f^{-1} = \text{id}_A$. Then $(*)$ is true.
 $f^{-1} \circ f = \text{id}_B$

Pf. Sup $f(x) = y$. Then $f^{-1}(f(x)) = f^{-1}(y)$.

Since $f^{-1} \circ f = \text{id}$, $f^{-1}(f(x)) = x$. Thus $x = f^{-1}(y)$.
 $\Leftarrow$ is similar. $\square$

